Group I Configurations

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Problem Statement I

For promoting its Travel offers to Privileged customers, a reputed Aviation Company decides to reach out to frequent air travelers through promotions. For this the company gets data feed from 3 sources through batch loads. The company assigns the task to Cognizant, to create and maintain a Master Hub of air travellers who may be Potential customers to the aviation company. Establish the intended Master Data Hub and generate the Golden copy of the customer base

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Informatica MDM Configurations:

Base Object

1. **Lookup Base Objects (Reference Data)**
   These tables store static reference data. They do not undergo match and merge.
   Table: C_LKP_GENDER

Attribute Name	Data Type	Key Type	Description / Purpose
ROWID_OBJECT	CHAR(14)	Primary Key	System-generated unique identifier.
GENDER_CD	CHAR(1)	Natural Key	The code ('M', 'F', 'U').
GENDER_DESC	VARCHAR(20)		Full description ('Male', 'Female', 'Unknown').

Table: C_LKP_COUNTRY

Attribute Name	Data Type	Key Type	Description / Purpose
ROWID_OBJECT	CHAR(14)	Primary Key	System-generated unique identifier.
COUNTRY_CD	CHAR(2)	Natural Key	ISO Country Code (e.g., 'US', 'IN').
COUNTRY_NAME	VARCHAR(50)		Full name of the country.

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2. **Parent Base Object**
   This is the core entity that defines a unique frequent flyer. Match rules will be configured heavily on this table to generate the Golden Copy.

Table: C_BO_CUSTOMER

Attribute Name	Data Type	Key/Trust	Description / Purpose
ROWID_OBJECT	CHAR(14)	Primary Key	The ultimate Golden ID for the frequent flyer.
FIRST_NAME	VARCHAR(50)	Trusted	Used in Exact and Fuzzy match rules (Person_Name).
LAST_NAME	VARCHAR(50)	Trusted	Used in Exact and Fuzzy match rules (Person_Name).

Attribute Name	Data Type	Key/Trust	Description / Purpose
DOB	DATE	Trusted	Used in Exact match rules. Validated for NULLs.
GENDER_CD	CHAR(1)	Foreign Key	Links to C_LKP_GENDER(GENDER_CD).
PASSPORT_NO	VARCHAR(20)	Trusted	Validated for length (<6 downgrade trust).

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3. **Child Base Objects**
   These tables hold the one-to-many relationships for the customer.

Table: C_BO_ADDRESS

Attribute Name	Data Type	Key/Trust	Description / Purpose
ROWID_OBJECT	CHAR(14)	Primary Key	System-generated unique ID for the address record.
CUST_ROWID	CHAR(14)	Foreign Key	Links to C_BO_CUSTOMER(ROWID_OBJECT).
ADDR_LINE_1	VARCHAR(100)		Primary street address. Used in Exact match.
ADDR_LINE_2	VARCHAR(100)		Nullable.
CITY	VARCHAR(50)		City of residence.
STATE	VARCHAR(50)		State of residence.
ZIP_CODE	VARCHAR(20)		Used in Exact match rules (cross-referenced by Parent).
COUNTRY_CD	CHAR(2)	Foreign Key	Links to C_LKP_COUNTRY(COUNTRY_CD).

Table: C_BO_CONTACT

Attribute Name	Data Type	Key/Trust	Description / Purpose
ROWID_OBJECT	CHAR(14)	Primary Key	System-generated unique ID for the contact record.

Attribute Name	Data Type	Key/Trust	Description / Purpose
CUST_ROWID	CHAR(14)	Foreign Key	Links to C_BO_CUSTOMER(ROWID_OBJECT).
EMAIL	VARCHAR(100)		Email address.
PHONE_NUMBER	VARCHAR(20)		Contact number (standardized in Staging).

Validation Rules

- **Validation Rule 1: Missing Date of Birth (DOB)**
  Aviation promotions heavily rely on age demographics. If a system provides a customer record without a Date of Birth, that record is considered incomplete and less reliable.
  - **Rule Name:** VAL_MISSING_DOB
  - **Target Base Object:** C_BO_CUSTOMER
  - **Validation Type:** Standard (SQL-based condition)
  - **Condition:** DOB IS NULL
  - **Action (Downgrade):** Downgrade Trust by **50%**.
  - **Target Columns for Downgrade:** FIRST_NAME, LAST_NAME, DOB
  - **Business Justification:** If a source system (like PRT or AGY) feeds a record without a DOB, downgrading its trust prevents that incomplete record from overriding a complete, high-quality record from the WEB source during the Match and Merge process.
- **Validation Rule 2: Invalid Passport Number Length**
  For an aviation company, the Passport Number is a highly critical identifier. Dummy data or placeholder values (like "123" or "NA") frequently enter through third-party agencies. A valid passport typically contains between 6 and 9 alphanumeric characters.
  - **Rule Name:** VAL_INVALID_PASSPORT_LEN
  - **Target Base Object:** C_BO_CUSTOMER
  - **Validation Type:** Standard (SQL-based condition)
  - **Condition:** LENGTH(PASSPORT_NO) < 6 OR PASSPORT_NO IS NULL
  - **Action (Downgrade):** Downgrade Trust by **60%**.
  - **Target Columns for Downgrade:** FIRST_NAME, LAST_NAME, PASSPORT_NO
  - **Business Justification:** Records with impossibly short or missing passport numbers indicate low data entry quality. A heavy 60% penalty ensures that this specific source row will almost certainly lose to a row that possesses a valid, full-length passport number.

Trust Scores

1. Parent Base Object: C_BO_CUSTOMER

Trust is critical here because this table holds the core identity of the frequent flyer. The validation rules we defined earlier (DOB and Passport) will actively downgrade these base scores if the data is poor.

Trusted Columns: FIRST_NAME, LAST_NAME, DOB, PASSPORT_NO

Source System	Initial Trust Score	Minimum Trust Score	Decay Period	Decay Type
WEB (Online)	**90**	85	12 Months	Linear
AGY (Agency)	**80**	70	12 Months	Linear
PRT (Partner)	**70**	55	12 Months	Linear

Business Logic: WEB data is entered directly by the customer, giving it the highest initial trust and lowest decay. PRT data comes from third-party partners; it starts with the lowest trust and decays rapidly as passenger details can change frequently without the partner knowing.

2. Child Base Object: C_BO_ADDRESS

Addresses change over time, so Trust and Decay are essential to ensure the most recent and reliable address wins.

Trusted Columns: ADDR_LINE_1, CITY, ZIP_CODE

Source System	Initial Trust Score	Minimum Trust Score	Decay Period	Decay Type
WEB (Online)	**90**	85	24 Months	Linear

Source System	Initial Trust Score	Minimum Trust Score	Decay Period	Decay Type
AGY (Agency)	**80**	70	24 Months	Linear
PRT (Partner)	**70**	55	24 Months	Linear

Business Logic: The WEB portal requires billing validation, making its addresses highly reliable (95). Decay is set over 24 months, as people move less frequently than they might change a phone number or email.

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3. Child Base Object: C_BO_CONTACT

Contact information becomes outdated very quickly. We apply an aggressive decay schedule here.

Trusted Columns: EMAIL, PHONE_NUMBER

Source System	Initial Trust Score	Minimum Trust Score	Decay Period	Decay Type
WEB (Online)	**95**	70	6 Months	Linear
AGY (Agency)	**80**	50	6 Months	Linear
PRT (Partner)	**65**	30	6 Months	Linear

Business Logic: An email or phone number provided 6 months ago to an airline partner (PRT) is highly suspect (drops to 30), whereas a direct update on the aviation company's WEB portal strongly overrides it.

Match/Merge Setup

1. Match Path Setup

To utilize a child column in a parent match rule (as required), you first establish a Match Path linking the parent and child Base Objects.

- **Path Name:** Path_Cust_to_Address
- **Direction:** C_BO_CUSTOMER (Parent) -> C_BO_ADDRESS (Child)
- **Join Condition:** C_BO_CUSTOMER.ROWID_OBJECT = C_BO_ADDRESS.CUST_ROWID

2. Match Columns Configuration

Before creating rules, you must define the Match Columns. These map the physical database columns to the tokens MDM uses for comparison.

At Parent BO (C_BO_CUSTOMER)

- **Person_Name (Fuzzy):** Mapped to FIRST_NAME and LAST_NAME.
    - *Purpose:* Handles typos and variations (e.g., "Johnathon Doe" vs. "John Doe").
- **Exact_DOB (Exact):** Mapped to DOB.
- **Exact_Passport (Exact):** Mapped to PASSPORT_NO.

At Child BO (C_BO_ADDRESS) *(Available to the Parent via Match Path)*

- **Exact_Zip (Exact):** Mapped to ZIP_CODE.
- **Exact_Address (Exact):** Mapped to ADDR_LINE_1.
- **Exact_Cust_Rowid (Exact):** Mapped to CUST_ROWID. (Ensures address consolidation only happens within the same parent profile).

3. Match Rules (Parent BO: C_BO_CUSTOMER)

The requirement mandates a minimum of two match rules at the parent level: one Exact and one Fuzzy using a child column.

Rule #	**Match Purpose**	**Rule Type**	**Match Columns Used**	**Auto / Manual Merge**
Rule 1	Strict Identity Match	**Exact**	Exact_Passport + Exact_DOB	**Auto-Merge**
Rule 2	Cross-System Variation Match	**Fuzzy**	Person_Name (Fuzzy) + Exact_Address_1**(Child Column via Match Path)**	**Manual Merge**

- **Business Logic for Rule 1:** If two systems provide the exact same Passport Number and Date of Birth, they are definitively the same frequent flyer. The system will automatically merge these records.
- **Business Logic for Rule 2:** *(Fulfills the specific project requirement).* If the name sounds similar (Fuzzy) and they reside at the exact same address line 1(Child Column), it is highly likely the same person. Because it relies on fuzzy logic, it queues the record for Manual Merge so a Data Steward can verify it in the UI before linking the profiles.

4. Match Rule (Child BO: C_BO_ADDRESS)

The requirement mandates a minimum of one match rule on a child BO to ensure duplicate child records (like two identical addresses from different source systems) are consolidated under the single golden parent.

Rule #	Match Purpose	Rule Type	Match Columns Used	Auto / Manual Merge
Rule 1	Address Consolidation	**Exact**	Exact_Address + Exact_Zip + CUST_ROWID	**Auto-Merge**

- **Business Logic:** If the WEB system and the AGY system both feed the address "123 Main St" with the zip code "10001" for the same customer, this rule consolidates those two address records into one Golden Address, preventing duplicate mailings for promotions.

Staging Tables

1. Customer Staging Tables (Parent)

You will create three separate staging tables mapped to the parent C_BO_CUSTOMER Base Object, all pulling from the common C_LND_CUSTOMER landing table.

- **Tables Created:**
  - C_STG_WEB_CUSTOMER (Filter applied: SRC_SYSTEM = 'WEB')
  - C_STG_AGY_CUSTOMER (Filter applied: SRC_SYSTEM = 'AGY')
  - C_STG_PRT_CUSTOMER (Filter applied: SRC_SYSTEM = 'PRT')

Staging Attributes & Mapping:

Staging Attribute	Data Type	Mapped From (Landing / Transformation)	Purpose in Staging
PKEY_SRC_OBJECT	VARCHAR(255)	Customer_ID (from CSV)	**Critical:** The unique identifier from the source system.
FIRST_NAME	VARCHAR(50)	FIRST_NAME (Cleansed)	Apply Title Case cleansing function here.
LAST_NAME	VARCHAR(50)	LAST_NAME (Cleansed)	Apply Title Case cleansing function here.
DOB	DATE	DOB	Passed through directly.

Staging Attribute	Data Type	Mapped From (Landing / Transformation)	Purpose in Staging
PASSPORT_NO	VARCHAR(20)	PASSPORT_NO (Cleansed)	Strip spaces and hyphens using string functions.
GENDER_CD	CHAR(1)	GENDER	Look up against C_LKP_GENDER to ensure validity.

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2. Address Staging Tables (Child)

You will create three staging tables mapped to the child C_BO_ADDRESS Base Object, pulling from C_LND_ADDRESS.

- **Tables Created:**
  - C_STG_WEB_ADDRESS (Filter applied: SRC_SYSTEM = 'WEB')
  - C_STG_AGY_ADDRESS (Filter applied: SRC_SYSTEM = 'AGY')
  - C_STG_PRT_ADDRESS (Filter applied: SRC_SYSTEM = 'PRT')

Staging Attributes & Mapping:

Staging Attribute	Data Type	Mapped From (Landing / Transformation)	Purpose in Staging
PKEY_SRC_OBJECT	VARCHAR(255)	ADDRESS_ID	Acts as the strictly unique primary key for the address record.

Staging Attribute	Data Type	Mapped From (Landing / Transformation)	Purpose in Staging
CUST_PKEY	VARCHAR(255)	Customer_ID	Used by MDM during the Load process to resolve the CUST_ROWID foreign key to the parent.
ADDR_LINE_1	VARCHAR(100)	ADDR_LINE_1	Passed through directly.
ADDR_LINE_2	VARCHAR(100)	ADDR_LINE_2	Passed through directly.
CITY	VARCHAR(50)	CITY (Cleansed)	Apply Title Case cleansing function.
STATE	VARCHAR(50)	STATE (Cleansed)	Apply Upper Case cleansing function to standardize state formats.
ZIP_CODE	VARCHAR(20)	ZIP_CODE	Passed through directly.

Staging Attribute	Data Type	Mapped From (Landing / Transformation)	Purpose in Staging
COUNTRY_CD	CHAR(2)	COUNTRY	Look up against C_LKP_COUNTRY.

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3. Contact Staging Tables (Child)

You will create three staging tables mapped to the child C_BO_CONTACT Base Object, pulling from C_LND_CONTACT.

- **Tables Created:**
  - C_STG_WEB_CONTACT (Filter applied: SRC_SYSTEM = 'WEB')
  - C_STG_AGY_CONTACT (Filter applied: SRC_SYSTEM = 'AGY')
  - C_STG_PRT_CONTACT (Filter applied: SRC_SYSTEM = 'PRT')

Staging Attributes & Mapping:

Staging Attribute	Data Type	Mapped From (Landing / Transformation)	Purpose in Staging
PKEY_SRC_OBJECT	VARCHAR(255)	CONTACT_ID	Acts as the strictly unique primary key for the contact record.

Staging Attribute	Data Type	Mapped From (Landing / Transformation)	Purpose in Staging
CUST_PKEY	VARCHAR(255)	Customer_ID	Resolves the foreign key to the Parent BO.
EMAIL	VARCHAR(100)	EMAIL (Cleansed)	Convert to lowercase, validate email format (@ symbol).
PHONE_NUMBER	VARCHAR(20)	PHONE_NUMBER (Cleansed)	Remove all non-numeric characters using Regex.

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4. Reference Data Staging Tables (Lookups)

Even though reference data is loaded via SQL insert scripts and doesn't participate in Match & Merge, it still requires Staging tables to move from Landing into the Base Objects. Because there is only one source (SYS_REF), you only need one staging table per Lookup BO.

- **Tables Created:**
    - C_STG_REF_GENDER (Filter applied: SRC_SYSTEM = 'SYS_REF')
    - C_STG_REF_COUNTRY (Filter applied: SRC_SYSTEM = 'SYS_REF')

Staging Attributes for Gender:

Staging Attribute	Data Type	Mapped From	Purpose in Staging
PKEY_SRC_OBJECT	VARCHAR(255)	GENDER_CD	The Code acts as the unique source key (e.g., 'M', 'F').
GENDER_CD	CHAR(1)	GENDER_CD	Passed through.
GENDER_DESC	VARCHAR(20)	GENDER_DESC	Passed through.

Staging Attributes for Country:

Staging Attribute	Data Type	Mapped From	Purpose in Staging
PKEY_SRC_OBJECT	VARCHAR(255)	COUNTRY_CD	The Code acts as the unique source key (e.g., 'US', 'IN').

Staging Attribute	Data Type	Mapped From	Purpose in Staging
COUNTRY_CD	CHAR(2)	COUNTRY_CD	Passed through.
COUNTRY_NAME	VARCHAR(50)	COUNTRY_NAME	Passed through.

Landing Tables

1. Parent Landing Table: C_LND_CUSTOMER

This table receives the core demographic data for the frequent flyer from the 3 CSV sources.

Attribute Name	Data Type	Description / Purpose
CUSTOMER_ID	VARCHAR(255)	The original Primary Key from the source CSV file.
FIRST_NAME	VARCHAR(50)	Customer's first name.
LAST_NAME	VARCHAR(50)	Customer's last name.
DOB	DATE	Date of Birth (standardized to DATE format by PowerCenter).
PASSPORT_NO	VARCHAR(20)	Customer's passport number.
GENDER	CHAR(1)	Source gender code (e.g., 'M', 'F') to be looked up in Staging.
SRC_SYSTEM	VARCHAR(20)	Target value: 'WEB', 'AGY', or 'PRT'.
LAST_UPDATE_DATE	DATE	Load timestamp.

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2. Child Landing Table: C_LND_ADDRESS

This table receives the physical location data. It must contain the CUSTOMER_ID to maintain the relationship to the parent record.

Attribute Name	Data Type	Description / Purpose
ADDRESS_ID	VARCHAR(255)	The unique identifier for the specific address record from the source system.
CUSTOMER_ID	VARCHAR(255)	Foreign Key linking back to C_LND_CUSTOMER.CUSTOMER_ID.
ADDR_LINE_1	VARCHAR(100)	Primary street address.
ADDR_LINE_2	VARCHAR(100)	Secondary street address (e.g., apartment, suite, or building number). Nullable.
CITY	VARCHAR(50)	City of residence.

Attribute Name	Data Type	Description / Purpose
STATE	VARCHAR(50)	State of residence.
ZIP_CODE	VARCHAR(20)	Postal code (Used later in the Parent Fuzzy Match Rule).
COUNTRY	CHAR(2)	Source country code (e.g., 'US', 'IN') to be looked up in Staging.
SRC_SYSTEM	VARCHAR(20)	Target value: 'WEB', 'AGY', or 'PRT'.
LAST_UPDATE_DATE	DATE	Load timestamp.

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3. Child Landing Table: C_LND_CONTACT

This table receives the communication data. Like the address table, it relies on the CUSTOMER_ID to link to the parent.

Attribute Name	Data Type	Description / Purpose
CONTACT_ID	VARCHAR(255)	The unique identifier for the specific contact record from the source system.
CUSTOMER_ID	VARCHAR(255)	Foreign Key linking back to C_LND_CUSTOMER.CUSTOMER_ID.
EMAIL	VARCHAR(100)	Customer's email address.
PHONE_NUMBER	VARCHAR(20)	Customer's contact number.
SRC_SYSTEM	VARCHAR(20)	Target value: 'WEB', 'AGY', or 'PRT'.
LAST_UPDATE_DATE	DATE	Load timestamp.

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4. Reference Landing Table: C_LND_GENDER

This table receives data strictly from your manual SQL insert scripts.

Attribute Name	Data Type	Description / Purpose
GENDER_CD	CHAR(1)	The code representing the gender (e.g., 'M').
GENDER_DESC	VARCHAR(20)	Full description (e.g., 'Male').

Attribute Name	Data Type	Description / Purpose
SRC_SYSTEM	VARCHAR(20)	Target value: 'SYS_REF' (Hardcoded in your insert script).
LAST_UPDATE_DATE	DATE	Load timestamp.

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5. Reference Landing Table: C_LND_COUNTRY

This table receives data strictly from your manual SQL insert scripts.

Attribute Name	Data Type	Description / Purpose
COUNTRY_CD	CHAR(2)	ISO Country Code (e.g., 'US').
COUNTRY_NAME	VARCHAR(50)	Full name of the country (e.g., 'United States').
SRC_SYSTEM	VARCHAR(20)	Target value: 'SYS_REF' (Hardcoded in your insert script).
LAST_UPDATE_DATE	DATE	Load timestamp.

Mappings

In Informatica MDM, while you have 1 Landing Table funnelling into 3 separate Staging Tables (for WEB, AGY, and PRT), the **column-to-column mapping logic remains exactly the same** across all three staging tables for a given Base Object. The only difference is the filter applied on the SRC_SYSTEM column during the stage job.

Here are the exact mappings for your documentation.

1. Customer Mapping (Parent)

Source: C_LND_CUSTOMER

Targets: C_STG_WEB_CUSTOMER, C_STG_AGY_CUSTOMER, C_STG_PRT_CUSTOMER

Landing Table Column (Left)	**MDM Hub Cleansing Logic (Optional/Recommended)**	**Staging Table Column (Right)**
CUSTOMER_ID	Direct Pass-through	**PKEY_SRC_OBJECT** *(Critical source key)*
FIRST_NAME	Title Case Function	FIRST_NAME
LAST_NAME	Title Case Function	LAST_NAME
DOB	Direct Pass-through	DOB
PASSPORT_NO	Strip non-alphanumeric characters	PASSPORT_NO
GENDER	Pass-through with Lookup Validation against C_STG_REF_GENDER	GENDER_CD

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2. Address Mapping (Child)

Source: C_LND_ADDRESS

Targets: C_STG_WEB_ADDRESS, C_STG_AGY_ADDRESS, C_STG_PRT_ADDRESS

Landing Table Column (Left)	**MDM Hub Cleansing Logic**	**Staging Table Column (Right)**
ADDRESS_ID	Direct Pass-through	**PKEY_SRC_OBJECT** *(Creates unique Address ID)*
CUSTOMER_ID	Direct Pass-through	**CUST_PKEY** *(Resolves Parent Foreign Key)*
ADDR_LINE_1	Direct Pass-through	ADDR_LINE_1
ADDR_LINE_2	Direct Pass-through	ADDR_LINE_2
CITY	Title Case Function	CITY
STATE	Apply Upper Case cleansing function	STATE
ZIP_CODE	Direct Pass-through	ZIP_CODE
COUNTRY	Pass-through with Lookup Validation against C_STG_REF_COUNTRY	COUNTRY_CD

3. Contact Mapping (Child)

Source: C_LND_CONTACT

Targets: C_STG_WEB_CONTACT, C_STG_AGY_CONTACT, C_STG_PRT_CONTACT

Landing Table Column (Left)	**MDM Hub Cleansing Logic**	**Staging Table Column (Right)**
CONTACT_ID	Direct Pass-through	**PKEY_SRC_OBJECT** *(Creates unique Contact ID)*
CUSTOMER_ID	Direct Pass-through	**CUST_PKEY** *(Resolves Parent Foreign Key)*
EMAIL	Lowercase Function	EMAIL
PHONE_NUMBER	Regex replace to format (e.g., (###) ###-####)	PHONE_NUMBER

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4. Reference Data Mapping (Lookups)

Because Reference Data uses a single source system (SYS_REF), it maps 1-to-1 from Landing to Staging without requiring multiple staging tables.

Gender Mapping

Source: C_LND_GENDER -> **Target:** C_STG_REF_GENDER

Landing Table Column (Left)	**MDM Hub Cleansing Logic**	**Staging Table Column (Right)**
GENDER_CD	Direct Pass-through	**PKEY_SRC_OBJECT**
GENDER_CD	Direct Pass-through	GENDER_CD

Landing Table Column (Left)	MDM Hub Cleansing Logic	Staging Table Column (Right)
GENDER_DESC	Direct Pass-through	GENDER_DESC

Country Mapping

Source: C_LND_COUNTRY -> **Target:** C_STG_REF_COUNTRY

Landing Table Column (Left)	MDM Hub Cleansing Logic	Staging Table Column (Right)
COUNTRY_CD	Direct Pass-through	**PKEY_SRC_OBJECT**
COUNTRY_CD	Direct Pass-through	COUNTRY_CD
COUNTRY_NAME	Direct Pass-through	COUNTRY_NAME

PowerCenter Configurations

Part 1: Source CSV File Definitions & Mock Data

To satisfy the requirement that *"Source Data has to be prepared in a way where both match & merge scenario & non-matched scenario will be covered,"* we will structure the 3 CSV files with deliberate data patterns.

File Structure (Common across all 3 files):

Customer_ID, Address_ID, Contact_ID, First_Name, Last_Name, DOB, Passport_No, Gender, Addr_Line_1, Addr_Line_2, City, State, Zip_Code, Country, Email, Phone_Number

1. File: WEB_Customers.csv (Online Bookings)

This file acts as our baseline high-trust data. (This is a sample data file)

Customer_ID	Address_ID	Contact_ID	First_Name	Last_Name	DOB	Passport_No	Gender	Addr_Line_1	Addr_Line_2	City	State	Zip_Code	Country	Email	Phone_Number
WEB_001	WEB_ADDR_001	WEB_CONT_001	John	Doe	05/15/1985	AB1234 56	M	123 Main St		New York	NY	10001	US	john.doe@email.com	555-0101
WEB_002	WEB_ADDR_002	WEB_CONT_002	Sarah	Connor	11/20/1990	XY987654	F	456 Elm St		Los Angeles	CA	90001	US	sarah.c@email.com	555-0202

- Scenario Note: WEB_002 will be our Non-Matched Scenario (a unique customer who only exists in the WEB system).

2. File: AGY_Customers.csv (Travel Agency)

This file contains variations of WEB_001 to test the Fuzzy Match Rule using the Child Zip Code. (This is a sample data file)

Customer_ID	Address_ID	Contact_ID	First_Name	Last_Name	DOB	Passport_No	Gender	Addr_Line_1	Addr_Line_2	City	State	Zip_Code	Country	Email	Phone_Number
AGY_998	AGY_ADDR_998	AGY_CONT_998	Jonathon	Doe	05/15/1985	AB1234 56	M	123 Main Street		New York	NY	10001	US	j.doe@agency.com	5550101

- Scenario Note: AGY_998 is a Match Scenario. The name is spelled "Jonathon" instead of "John" (Fuzzy Name Match), but the Zip_Code "10001" is an exact match. This will trigger Parent Match Rule 2 (Fuzzy + Child Column) and queue for Manual Merge.

3. File: PRT_Customers.csv (Airline Partners)

This file contains incomplete data to test the Validation Rules (Trust Downgrade) and the Exact Match Rule. (This is a sample data file)

Customer_ID	Address_ID	Contact_ID	First_Name	Last_Name	DOB	Passport_No	Gender	Addr_Line_1	Addr_Line_2	City	State	Zip_Code	Country	Email	Phone_Number
PRT_X12	PRT_ADDR_X12	PRT_CONT_X12	John	Doe			M	123 Main St		NYC	NY	10001	US		

- Scenario Note: PRT_X12 is missing a DOB and Passport. This triggers the Validation Rules to heavily downgrade its trust. Because it matches the Exact Name and Exact

Zip Code, MDM will merge it with WEB_001, but the WEB system's data will win the Golden Copy due to the trust downgrade on the PRT record.

Part 2: PowerCenter Mapping Specifications

You will create 3 mappings. The logic for all three is identical, with the only difference being the hardcoded string for the SRC_SYSTEM port.

Mapping Architecture Pattern: 1 Flat File Source -> 1 Expression Transformation -> 1 Router Transformation -> 3 Relational Targets.

Mapping 1: m_WEB_to_LND_CUSTOMER

A. Source Definition & Source Qualifier (SQ_WEB_Customers)

- Reads the CSV file as string data types.

B. Expression Transformation (EXP_Standardize_and_Validate)

- Pulls all ports from the Source Qualifier.
- Applies basic data type conversions and generates the critical MDM system columns.
- Generates validation flags (O_IS_VALID_ROW) to identify records with correct email and phone number formats.

Input Port (From SQ)	Output Port Name (To Targets)	Expression Logic (Transformation)	Data Type
Address_ID	O_ADDRESS_ID	LTRIM(RTRIM(Address_ID))	String
Customer_ID	O_CUSTOMER_ID	LTRIM(RTRIM(Customer_ID))	String
First_Name	O_FIRST_NAME	UPPER(LTRIM(RTRIM(First_Name)))	String
Last_Name	O_LAST_NAME	UPPER(LTRIM(RTRIM(Last_Name)))	String

Input Port (From SQ)	Output Port Name (To Targets)	Expression Logic (Transformation)	Data Type
DOB	O_DOB	TO_DATE(DOB, 'MM/DD/YYYY')	Date/Time
Passport_No	O_PASSPORT_NO	UPPER(LTRIM(RTRIM(Passport_No)))	String
Gender	O_GENDER	UPPER(LTRIM(RTRIM(Gender)))	String
Addr_Line_1	O_ADDR_LINE_1	LTRIM(RTRIM(Addr_Line_1))	String
Addr_Line_2	O_ADDR_LINE_2	LTRIM(RTRIM(Addr_Line_2))	String
City	O_CITY	UPPER(LTRIM(RTRIM(City)))	String
State	O_STATE	UPPER(LTRIM(RTRIM(State)))	String
Zip_Code	O_ZIP_CODE	LTRIM(RTRIM(Zip_Code))	String
Country	O_COUNTRY	UPPER(LTRIM(RTRIM(Country)))	String
Contact_ID	O_CONTACT_ID	LTRIM(RTRIM(Contact_ID))	String
Email	O_EMAIL	LOWER(LTRIM(RTRIM(Email)))	String

Input Port (From SQ)	Output Port Name (To Targets)	Expression Logic (Transformation)	Data Type
Phone_Number	O_PHONE_NUMBER	LTRIM(RTRIM(Phone_Number))	String
(None)	V_CLEAN_PHONE (Variable)	REG_REPLACE(O_PHONE_NUMBER, '[^0-9]', '')	String
(None)	V_IS_NULL_EMAIL (Variable)	IIF(ISNULL(O_EMAIL) OR LENGTH(O_EMAIL) = 0, 1, 0)	Integer
(None)	V_IS_VALID_EMAIL (Variable)	IIF(REG_MATCH(O_EMAIL, '^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$'), 1, 0)	Integer
(None)	V_IS_VALID_PHONE (Variable)	IIF(NOT ISNULL(O_PHONE_NUMBER) AND LENGTH(V_CLEAN_PHONE) >= 10, 1, 0)	Integer
(None)	O_IS_VALID_ROW (Output)	IIF(V_IS_VALID_PHONE = 1 AND (V_IS_VALID_EMAIL = 1 OR V_IS_NULL_EMAIL = 1), 1, 0)	Integer
(None)	O_SRC_SYSTEM	'WEB'	String
(None)	O_LAST_UPDATE	SYSDATE	Date/Time

C. Connecting the Expression to the Router (EXP_Standardize_and_Validate -> RTR_Quality_Check)

We will drag only the Output (O_) ports from the Expression Transformation into the Input ports of the Router Transformation. (Variable ports like V_CLEAN_PHONE remain internal to the Expression and are not passed forward).

Map the following ports directly:

- O_ADDRESS_ID -> IN_ADDRESS_ID
- O_CUSTOMER_ID -> IN_CUSTOMER_ID
- O_FIRST_NAME -> IN_FIRST_NAME
- O_LAST_NAME -> IN_LAST_NAME
- O_DOB -> IN_DOB
- O_PASSPORT_NO -> IN_PASSPORT_NO
- O_GENDER -> IN_GENDER
- O_ADDR_LINE_1 -> IN_ADDR_LINE_1
- O_ADDR_LINE_2 -> IN_ADDR_LINE_2
- O_CITY -> IN_CITY
- O_STATE -> IN_STATE
- O_ZIP_CODE -> IN_ZIP_CODE
- O_COUNTRY -> IN_COUNTRY
- O_CONTACT_ID -> IN_CONTACT_ID
- O_EMAIL -> IN_EMAIL
- O_PHONE_NUMBER -> IN_PHONE_NUMBER
- O_SRC_SYSTEM -> IN_SRC_SYSTEM
- O_LAST_UPDATE -> IN_LAST_UPDATE
- O_IS_VALID_ROW -> IN_IS_VALID_ROW

D. Router Transformation (RTR_Quality_Check)

- Purpose: Filters out bad data based on the validation flags generated in the Expression transformation.
- Groups:
    - Valid_Records: Condition set to IN_IS_VALID_ROW = 1. Ports from this group are connected to the Landing Tables.
    - Default: Captures invalid data (IN_IS_VALID_ROW = 0) and drops the records from the flow.

E. Target Connections (Linking the Router to the 3 Landing Tables)

From the Valid_Records group of the RTR_Quality_Check transformation, you will drag the respective output ports to the three target tables simultaneously.

- Link to Target 1: C_LND_CUSTOMER
    - IN_CUSTOMER_ID -> CUSTOMER_ID
    - IN_FIRST_NAME -> FIRST_NAME
    - IN_LAST_NAME -> LAST_NAME
    - IN_DOB -> DOB
    - IN_PASSPORT_NO -> PASSPORT_NO
    - IN_GENDER -> GENDER
    - IN_SRC_SYSTEM -> SRC_SYSTEM
    - IN_LAST_UPDATE -> LAST_UPDATE_DATE
- Link to Target 2: C_LND_ADDRESS
    - IN_ADDRESS_ID -> ADDRESS_ID *(Primary Key for Address)*
    - IN_CUSTOMER_ID -> CUSTOMER_ID *(Acts as the Foreign Key)*
    - IN_ADDR_LINE_1 -> ADDR_LINE_1
    - IN_ADDR_LINE_2 -> ADDR_LINE_2
    - IN_CITY -> CITY
    - IN_STATE -> STATE
    - IN_ZIP_CODE -> ZIP_CODE
    - IN_COUNTRY -> COUNTRY
    - IN_SRC_SYSTEM -> SRC_SYSTEM
    - IN_LAST_UPDATE -> LAST_UPDATE_DATE
- Link to Target 3: C_LND_CONTACT
    - IN_CONTACT_ID -> CONTACT_ID *(Primary Key for Contact)*
    - IN_CUSTOMER_ID -> CUSTOMER_ID *(Acts as the Foreign Key)*
    - IN_EMAIL -> EMAIL
    - IN_PHONE_NUMBER -> PHONE_NUMBER
    - IN_SRC_SYSTEM -> SRC_SYSTEM

- IN_LAST_UPDATE -> LAST_UPDATE_DATE

Mapping 2 & 3: m_AGY_to_LND_CUSTOMER & m_PRT_to_LND_CUSTOMER

You can simply copy the m_WEB_to_LND_CUSTOMER mapping two times to create the mappings for AGY and PRT.

The only required changes inside the copied mappings are:

1. Source Qualifier: Point them to AGY_Customers.csv and PRT_Customers.csv respectively.
2. Expression Transformation: Change the hardcoded O_SRC_SYSTEM output port from 'WEB' to 'AGY' and 'PRT'.